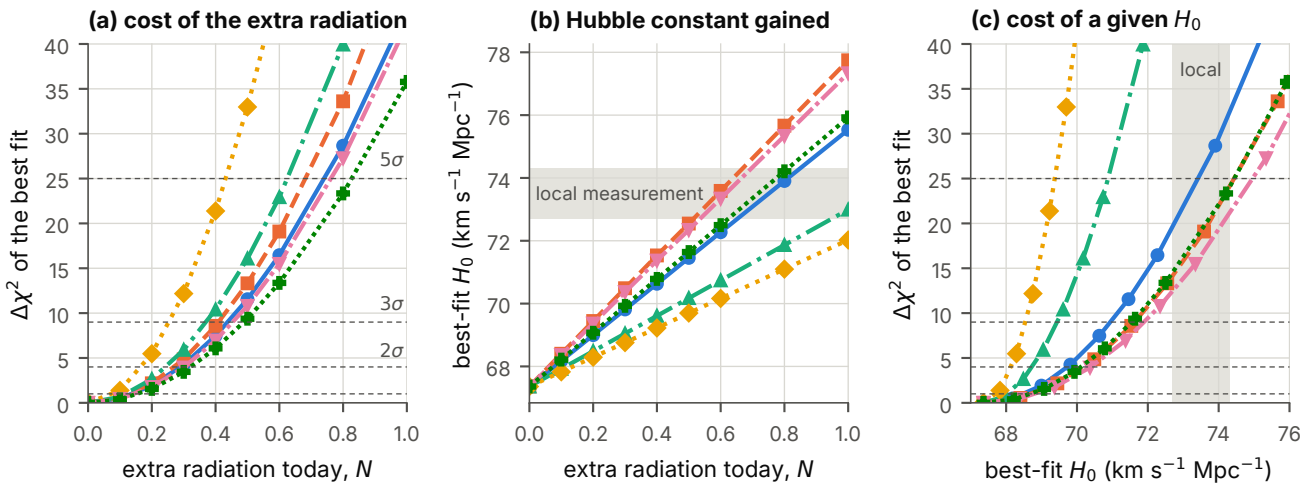
(a) cost of the extra radiation
$\Delta\chi^2$ of the best fit
extra radiation today, N
5σ
3σ
2σ
(b) Hubble constant gained
best-fit H_0 (km s^{-1} Mpc^{-1})
local measurement
extra radiation today, N
(c) cost of a given H_0
$\Delta\chi^2$ of the best fit
local
best-fit H_0 (km s^{-1} Mpc^{-1})
free-streaming radiation
fluid, no step
model fluid, $z_t = 10^4$
model fluid, $z_t = 2 \times 10^4$
model fluid, $z_t = 10^6$
Wess-Zumino fluid, $z_t = 2 \times 10^4$